

Article

Enhancing the Effectiveness of Recognition Memory in a Virtual Memory Palace Using Worlds-in-Miniature

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Abstract: The memory palace, a mnemonic technique from ancient Greece, remains widely used for memory competitions and learning. This study introduces a procedurally generated virtual memory palace combined with the Worlds-in-Miniature (WIM) interface in virtual reality. WIM enhances spatial perception by providing a miniaturized visualization of the memory palace and enabling instant movement between locations. Focusing on recognition memory, an experiment with 40 participants was conducted to measure memory performance under conditions with and without the WIM interface. Each participant memorized 30 target images within a procedurally generated memory palace and completed a recognition task distinguishing targets from non-targets. Data included immediate recognition performance and follow-up assessments at 7 and 14 days to evaluate long-term retention. The results showed that the WIM interface significantly improved recognition memory, particularly among participants with lower initial success rates. By providing spatial structuring, WIM facilitated stronger memory associations, leading to enhanced recall accuracy and sustained long-term retention. This method demonstrates the potential of VR memory palaces as effective tools for supporting memorization-based learning in education and training contexts.

Keywords: memory palace; Worlds-in-Miniature; virtual reality



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1. Introduction

The memory palace, also known as the method of loci, is a mnemonic technique that originated in ancient Greece and Rome [1]. It involves visualizing a specific location or path in one's mind and 'placing' the information to be remembered within that space. By utilizing spatial thinking, this method enhances memory [2]. Once a virtual space is created in the mind and the information is placed, the individual can easily recall the information by mentally retracing a specific path within that space. This technique is still widely used today, especially in situations like memory competitions, where large amounts of information need to be memorized in a short amount of time. However, effectively using the memory palace requires considerable training, making it difficult for the average person to access easily [3].

Researchers in computer graphics and virtual reality have developed virtual memory palace methods that allow even untrained individuals to experience and utilize the memory-enhancing effects of the memory palace [4]. Instead of mentally visualizing and exploring a location, virtual techniques involve physically viewing the space through graphic displays and interacting via human-computer interfaces. Initially, these methods were implemented in desktop PC (personal computer) environments, using three-dimensional (3D) rendering

and interaction through keyboards or mice. More recently, virtual environments have been visualized using VR head-mounted displays (HMDs), with interaction facilitated by 3D input devices such as motion controllers, providing a more immersive experience [5]. Comparative studies have shown that VR-implemented memory palaces are highly effective in improving memory retention [6].

Despite their potential, existing virtual memory palaces rely heavily on pre-designed spaces where users either place information arbitrarily or interact with pre-positioned information [7]. This pre-design limits flexibility compared to the original memory palace, where users could freely imagine a familiar space. Familiar environments, in traditional memory palaces, enhance memorization by leveraging the user's prior knowledge. In contrast, pre-designed virtual spaces often introduce unfamiliarity, reducing their effectiveness. Addressing this limitation, our study proposes a procedurally generated virtual memory palace that allows users to create new spaces in real-time. This approach offers flexibility and theoretically enables an infinite number of unique spaces for memorization. Since the generated spaces are unfamiliar to users, our design incorporates an intuitive interface to help them quickly understand and navigate these new environments.

To support spatial awareness and improve memorization, we utilized the Worlds-in-Miniature (WIM) interface [8]. Traditionally used in virtual reality for spatial navigation, WIM provides users with a global third-person perspective by visualizing a miniaturized version of the virtual environment. Users can see their position and orientation through an avatar, complementing their local first-person view within VR. This dual perspective enables users to effectively encode spatial information, even in unfamiliar environments, and enhances both short-term and long-term memory retention.

By integrating real-time spatial generation and the WIM interface, our study aims to overcome the limitations of pre-designed virtual memory palaces and provide a user-friendly, flexible solution for memorization-based tasks [9,10].

2. Related Work

2.1. Memory Palaces

The historical and philosophical development of the memory palace is well documented in *The Art of Memory* by Frances Yates [1]. The origins of the memory palace date back to the fifth century BCE, with its earliest recorded use attributed to the Greek poet Simonides. Memory formation and retrieval are closely tied to spatial context, as shown in studies emphasizing the primacy of spatial information in organizing event memories [1,2,9–12]. According to legend, Simonides used his memory of the layout of a banquet hall to identify the bodies of guests after a catastrophic building collapse. This experience led him to develop the memory palace as a method for improving memory. The technique was later adopted and systematized by notable figures such as Cicero, who utilized it extensively in rhetoric and public speaking [3]. Today, the memory palace remains a vital mnemonic device, widely used in memory competitions and educational contexts where the retention of large amounts of information is essential [6].

The cognitive and neurological basis of the memory palace is supported by extensive research in neuroscience. Studies have shown that spatial processing and memory formation are closely linked through the activity of the hippocampus, a brain region critical for encoding and retrieving memories [2,13]. The hippocampus contains specialized neurons known as place cells, which activate when an individual is in a specific location, and grid cells, which create a coordinate-like system to track movement through space [14]. These cells work together to form a mental map of one's surroundings, enabling effective spatial navigation and memory association. Experimental studies have demonstrated that individuals trained in the memory palace technique exhibit exceptional spatial memory

and engage brain regions associated with spatial processing. For example, research using MRI scans of London taxi drivers, who must memorize the city's complex street network, revealed that they had significantly larger hippocampi than those of the general population [15]. This suggests that extensive spatial training can lead to measurable structural changes in the brain.

Recent studies have extended this understanding to practical applications of the memory palace. Virtual memory palaces, implemented through computer-generated environments, provide a structured way for users to harness the benefits of spatial memory [7]. Unlike traditional memory palaces, virtual implementations allow for visual interaction with pre-designed spaces or dynamically generated environments. However, current research highlights several limitations, such as the reduced flexibility of pre-designed environments and the additional cognitive load required to familiarize users with unfamiliar spaces [4]. This has led to recent efforts to improve the adaptability and usability of virtual memory palaces through real-time spatial generation and enhanced spatial navigation interfaces, which have shown promising results in improving memory retention [6].

By leveraging this connection between spatial cognition and memory, our study aims to address the limitations of existing virtual memory palaces. Specifically, we focus on overcoming the challenges associated with unfamiliar environments through procedural generation and intuitive spatial interfaces, such as the Worlds-in-Miniature (WIM) system [8]. Our approach aims to enhance both short-term and long-term memory retention by enabling users to intuitively recognize and navigate newly generated spaces.

2.2. Virtual Memory Palaces

The original memory palace technique is widely accessible and can be learned quickly. However, effectively utilizing it for storing and recalling large volumes of information often requires structured training, making its full potential challenging to access for untrained users. Researchers have explored whether the use of computer graphics technology could enable the general public to create virtual memory palaces, visually arranging the information they want to remember in such a way that it yields memory enhancement effects comparable to those of traditional memory palaces. Harman was an early pioneer in proposing the virtualization of memory palaces using computers [4]. Fassbender and Heiden attempted to create virtual memory palaces as a substitute for traditional ones [16], and Legge et al. demonstrated that virtual memory palaces could provide memory enhancement effects on par with those of traditional memory palaces [17]. Similarly, Reggente et al. showed that utilizing the method of loci within virtual reality environments improves memory performance by explicitly binding objects to spatial contexts, thereby enhancing subsequent memory recall [12,18]. An interesting example of a virtual memory palace is the work of Das et al., who proposed using a 3D-rendered virtual maze (the memory palace) on a mobile phone, with the path taken through the maze serving as a password [19]. They confirmed that this approach was easier to remember than the traditional 9-dot password system. These early virtual memory palaces were typically implemented based on images rendered with 3D graphics on desktop monitors. However, numerous studies have shown that immersive environments experienced with virtual reality (VR) headsets are more effective for memory retention than desktop environments [5].

Subsequent studies have explored methods to engage in virtual memory palaces in an immersive way using VR headsets. Krokos et al. found that providing a VR version of the memory palace led to significantly improved recall abilities, compared to the desktop version [6]. Peeters and Segundo-Ortin showed that an embodied approach using VR can help to understand the utility of the memory palace and further proposed guidelines for effectively designing virtual memory palaces [20]. Yang et al. confirmed that applying

VR memory palaces in academic recall and recognition experiments was more effective than traditional memorization or image-based memory palaces [7]. Moll and Sykes experimentally demonstrated that VR memory palaces allowed participants to recall more words compared to simple memorization, and that even short training periods could optimize the learning of the memory palace technique [21]. Building on these studies, this article proposes a new method for VR memory palaces that enhances spatial perception and movement abilities, offering a more advanced approach compared to previous VR memory palaces.

2.3. Spatial Perception and Navigation in VR

Spatial perception and movement in the real world differ significantly from those in mentally created spaces; however, in virtual reality (VR), they can be designed to resemble the cognitive processes of mental navigation. In mentally created spaces, individuals can freely adjust various aspects such as first-person/third-person viewpoints, macroscopic/microscopic perspectives, continuous/discrete movement, and dynamic navigation through imagined structures. However, research on spatial perception and movement in VR has primarily focused on limited and often rigid navigation systems, typically offering either macroscopic or microscopic perspectives through a single-level first-person view.

Krekhov et al. proposed a technique in which the user's avatar size dynamically changes, enabling users to transition between large-scale exploration and precise microscopic interaction through continuous movement [22]. Abtahi et al. demonstrated that, by increasing virtual walking speeds beyond real-world limits, users could navigate large-scale virtual environments efficiently without breaking immersion [23]. Weissker et al. employed discrete teleportation mechanisms to switch between macroscopic and microscopic navigation scales, making spatial exploration flexible and intuitive [24]. Lee et al. proposed a method in which the user's avatar in virtual reality automatically adjusts the scale of the virtual environment based on the size of the object being manipulated. This allows for continuous transitions between macroscopic and microscopic perspectives, enabling the convenient handling of 3D objects [25]. However, these approaches primarily focus on mobility and immersion without addressing the spatial structuring required for effective memory encoding and retrieval.

The WIM method proposed by Stoakley et al. offers an effective solution by integrating both global and local spatial perspectives [8]. By presenting a miniature version of the virtual environment, WIM enables users to utilize both first-person and third-person perspectives simultaneously. Users can continuously navigate through the space from a first-person perspective while manipulating the miniature for macroscopic movements. This dual-perspective system significantly enhances spatial awareness and navigation flexibility compared to traditional VR navigation techniques [26,27].

WIM has evolved into various forms, such as SSWIM (supporting scaling and scrolling within the miniature) [26], SWIM (providing object-specific scaling) [27], and Step WIM (enabling body-driven navigation without hand input) [28]. Danyluk et al. identified key design elements for WIM interfaces that emphasize multi-dimensional spatial control and proposed future applications in cognitive and spatial learning [29].

Navigation plays a critical role in memory encoding and retrieval within virtual memory palaces, as spatial navigation directly influences how effectively users can organize and recall information [12,17]. Studies demonstrate that flexible navigation systems, particularly those offering dynamic control over movement and multi-scale perspectives, enhance both short-term recall and long-term memory retention [6,20]. The integration of WIM into a virtual memory palace is designed to harness these navigation benefits, allowing

users to build stronger spatial associations and improve their ability to encode and retrieve information effectively.

3. Pilot Study

As a pilot study, we explored various methods to implement a virtual memory palace and tested its feasibility with a total of nine participants across two experiments. To evaluate the effectiveness of creating and memorizing in a virtual memory palace, we conducted a pilot test without requiring prior knowledge of the memory palace technique. In this pilot test, we explored two approaches: comparing memorization in randomly generated environments with rote memorization, and introducing gamification to foster spatial familiarity. The first approach demonstrated that the unfamiliarity of the randomly generated environments reduced memory performance, while the second approach aimed to address this limitation by enhancing the user's engagement with the virtual space. The findings obtained from these two studies underscored the importance of designing an interface that allowed users to intuitively understand and utilize virtual spaces, forming the basis for the improved approach proposed in this study. The virtual environment for this pilot study was developed using Unity Engine and experienced using a Meta Quest 3 VR headset.

3.1. Pilot Study 1—Simple Memory Palaces

The first pilot experiment investigated the differences in memorization performance between rote memorization and the virtual memory palace. The virtual environment was developed using Unity Engine and experienced via a Meta Quest 3 VR headset. The virtual memory palace consisted of 10 rooms randomly placed within an open 3D space, with road connections between the rooms. Each room contained eight unique symbols randomly selected from a pool of approximately 2000 icons. The world was designed to limit spatial dispersion by ensuring minimum and maximum distances between rooms (see Figure 1a).

Participants explored the virtual memory palace for 5 min, during which symbols were displayed in fixed locations within the rooms. Participants had no time limit for observing each symbol and could spend as much time as needed to memorize them before moving to the next location. The symbols were observed through the VR HMD in a first-person perspective, simulating a natural exploration experience (Figure 1b). In contrast, participants in the rote memorization condition observed symbols statically displayed on a 2D screen within the VR environment without moving (Figure 1c). The eight symbols were displayed simultaneously on a static whiteboard screen, and participants could navigate between different sets of symbols using UI (user interface) controls, specifically by pressing left or right buttons to flip through pages.

During the memory palace condition, participants physically navigated through the virtual space using VR controllers to move between rooms and observe symbols. In the rote memorization condition, participants remained stationary and observed symbols sequentially on the 2D display. A total of six participants, recruited from the university's graduate program, participated in both conditions for 5 min each. Immediately after each task, participants were tested on their ability to recall and recognize symbols by identifying whether a given symbol was part of the target set or not.

The results of this pilot study showed that participants using the virtual memory palace achieved a memorization accuracy of 90%, compared to 85% for rote memorization. The virtual memory palace demonstrated slightly higher memorization performance. However, participants reported that exploring the unfamiliar virtual memory palace caused significant fatigue and was perceived as a waste of time. Additionally, they reported that the open spaces between rooms caused confusion and dizziness.

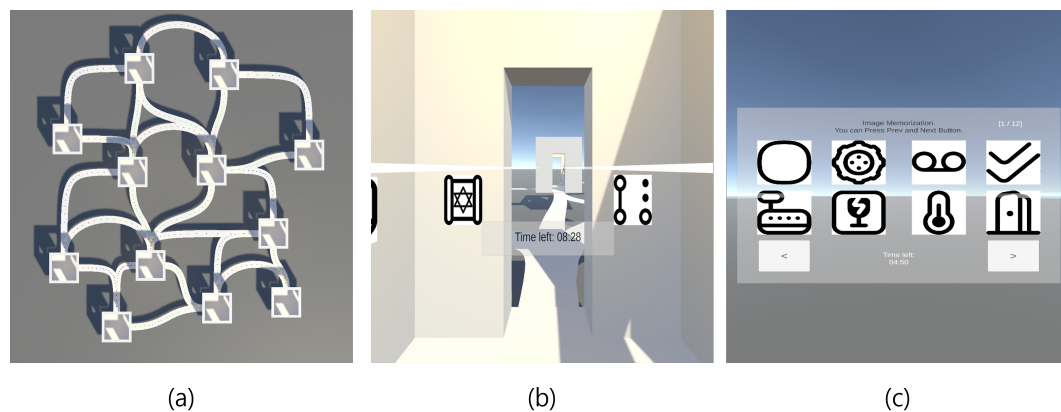


Figure 1. A simple memory palace for the first preliminary study: (a) An aerial view of the entire memory palace. (b) A first-person view of a room. (c) A display board for rote memorization.

3.2. Pilot Study 2—Gamified Memory Palaces

In the pilot study 1, we found that familiarity is a key factor in reducing cognitive load during navigation and enhancing memory retention. To address the challenge of unfamiliar spaces, we implemented a gamified memory palace, structured as an indoor dungeon and consisting of rooms and corridors. This environment was designed to increase engagement and foster familiarity through repeated interactions.

The virtual memory palace consisted of 10 rooms randomly generated within a predefined size range, connected by corridors. The configuration included one starting room, eight rooms containing collectible coins, and one room containing a box (Figure 2). Participants navigated through this virtual space using VR controllers, physically moving through corridors and rooms to complete tasks.

Participants first played a 10-minute mini-game, where they collected coins scattered across the rooms and deposited them into the box. The objective of the game was designed to encourage participants to revisit different rooms multiple times, increasing spatial familiarity. After the game, 10 images were placed in different rooms, with each room containing a unique symbol in a fixed position. Participants were instructed to memorize the symbols and their locations. During this phase, participants experienced the memory palace from a first-person perspective through the VR HMD.

After completing the memorization phase, participants revisited the virtual memory palace to take the memory test. Each room contained an interactive whiteboard displaying multiple symbols, and participants used buttons on the whiteboard to select the correct symbol corresponding to the room's location. This interaction required participants to recall both the spatial layout and the specific symbols associated with each room.

As a comparative condition, participants explored the same virtual environment freely for 10 min without any specific tasks or game mechanics. Instead of collecting coins or achieving goals, participants simply navigated through the rooms and corridors to observe the symbols. Once the exploration ended, the same memorization task was conducted, and participants took the memory test in the same way as in the gamified condition.

The pilot study involved three participants, who completed both the gamified and non-gamified conditions. The memory retention rate was 93% in the gamified condition and 80% in the non-gamified condition, suggesting a potential benefit of gamification in memory retention. However, given the small sample size, these findings should be interpreted as exploratory, and additional studies with larger participant groups are necessary to validate the observed effects.

Participants reported that the gamified condition made the exploration process more engaging and enjoyable, which may have contributed to better memorization performance.

However, participants also noted that the additional time required for familiarization with the environment could be a limitation when applied to real-world scenarios where rapid memorization is necessary. Future improvements could focus on reducing the time required for spatial familiarization while maintaining the benefits of gamification.

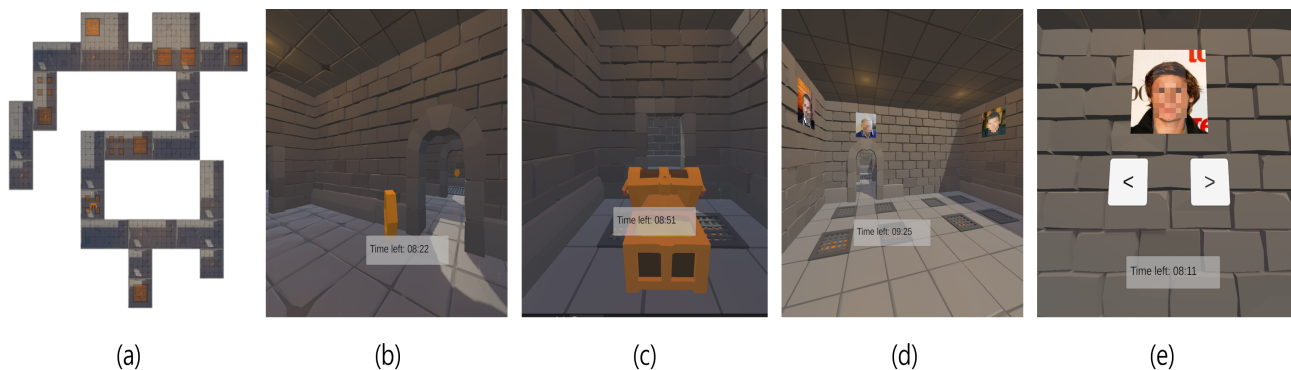


Figure 2. Gamified virtual memory palace: (a) An aerial view of the entire memory palace. (b) A collectable coin found in a room. (c) A box for keeping coins. (d) Target images on the walls for the user to memorize. (e) Test screen for checking whether a certain image has been included in the memory palace or not. (The faces of the characters have been blurred to prevent infringement of portrait rights).

4. Materials and Methods

This section covers the overall aspects of the experiment, including the hypotheses about the effectiveness of the proposed virtual memory palace, a description of the participant groups, the experimental procedure, and the data collection methods.

4.1. Virtual Memory Palace

Our virtual memory palace was procedurally generated by dynamically creating virtual worlds with rooms of various sizes arranged on a 2D grid-based tile map (Figure 3). The rooms were connected via paths, ensuring navigability, and Prim's Minimum Spanning Tree algorithm was applied to maintain connectivity. Each room displayed memorization targets on walls, with image placement optimized to reduce visual overload. The system was scalable for different experimental configurations but consistently used 30 images for this study.

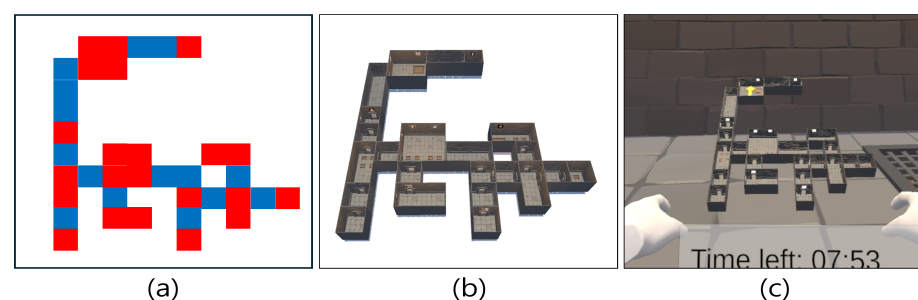


Figure 3. Procedurally generated virtual memory palace using Worlds-in-Miniature: (a) A procedurally generated 2D tile map (room and path are colored red and blue, respectively). (b) A 3D environment, made by replacing 2D tiles with 3D objects. (c) A World-in-Miniature shown in front of the user equipped with VR HMD, in which the yellow arrow represents user's position and rotation.

To enhance spatial navigation, we implemented the Worlds-in-Miniature (WIM) interface, which provided users with a synchronized miniature view of the virtual memory palace. The WIM supported two interaction modes: a fixed mode offering a stable overview

of the environment and a dynamic mode that synchronized with the user's viewpoint. Users could directly manipulate their position via the WIM by dragging an avatar to a new location, allowing for efficient navigation without the need for physical exploration. The actual implementation of the virtual memory palace is described in detail in Appendix A.

4.2. Hypotheses

Before conducting the experiment, we established hypotheses regarding the expected outcomes, as follows:

- H1: In memory evaluations, users provided with WIM will achieve higher scores in both short-term and long-term memory, compared to those without WIM.
- H2: Participants with inferior memorization ability will demonstrate significant differences in recognition performance between the conditions with and without WIM compared to those with superior memorization ability.
- H3: Participants will feel more at ease and encounter fewer instances of confusion when using WIM.

These hypotheses provide a structured framework for assessing how the presence or absence of WIM impacts user behavior and memory performance in virtual environments.

4.3. Participants

The participants comprised 40 undergraduate and graduate students, including 26 men and 14 women, with an average age of 23 years. Recruitment included questions about their experience with 3D games and VR. For 3D games, experience ranged from no experience (0 points) to playing for over 6 h (3 points). VR experience ranged from never used (0 points) to regular use (3 points). The average score for 3D games was 1.6, with six participants having no experience. Most participants had some game experience but were not hardcore players. For VR, the average was 1.5, with five participants having no experience. The rest were generally experienced users.

Participants were provided with a detailed explanation of the experiment and asked to sign a consent form. Following this, a pre-test was conducted to fairly divide them into two memorization groups based on their memorization abilities. The pre-test involved memorizing the sequence of 10 randomly drawn cards from a deck within one minute. This test was designed to ensure that participants with varying levels of memorization skills were evenly distributed across both groups. The average score was 6.5, with the highest being 10 and the lowest 3. Based on the pre-test results and survey data, participants were divided into Group 1 and Group 2.

To ensure fairness and minimize potential biases, all participants were required to complete both the experiment with WIM and the experiment without WIM. However, to control for possible advantages in the second experiment due to environmental adaptation or increased familiarity with the controller, we made Group 1 and Group 2 perform the with-WIM and without-WIM conditions in opposite orders. Specifically, Group 1 completed the experiment with WIM first, followed by the experiment without WIM. Conversely, Group 2 completed the experiment without WIM first, and then performed the experiment with WIM. This process was designed to minimize potential biases while ensuring a balanced experimental design.

4.4. Dataset for Memorization

In our experiment, the memorization targets were based on the CelebA dataset to provide a diverse set of easily distinguishable images, making it suitable for memorization experiments [30]. This dataset consists of over 200,000 celebrity images collected from the

Internet. The dataset was divided randomly into four subsets for memorization evaluation. Each of these subsets, labeled A, B, C, and D approximately includes 50,000 images.

Set A was used in the experiment with WIM, where 30 random images were extracted for each participant. Set B was used in the experiment without WIM, with 30 random images extracted again. Sets C and D served as dummy data during the memory evaluation, with 30 random images extracted from each. After the experiment with WIM, set C was combined and shuffled with the 30 images from set A, creating a total of 60 images for participants to judge as true or false regarding their inclusion in the virtual memory palace. Similarly, after the experiment without WIM, set D was combined with set B.

The reason we divided the data into four subsets was not only to ensure a balanced and randomized distribution of memorization targets but also to prevent any overlap between the targets and dummy data across the two experimental conditions. This approach allowed us to maintain consistency while minimizing potential biases in the evaluation process.

4.5. Procedure

Participants experienced the virtual memory palace with both modes of the WIM interface during their exploration: the fixed mode and the dynamic mode. The fixed mode provided a stable miniature view of the environment, while the dynamic mode synchronized the miniature with the user's viewpoint. All participants used both modes sequentially within the same experimental session, and no mode-specific comparisons were conducted. Instead, the results reflect the combined influence of both WIM modes on memory performance. The experiment was divided into two parts: short-term memory evaluation and long-term memory evaluation (Figure 4).

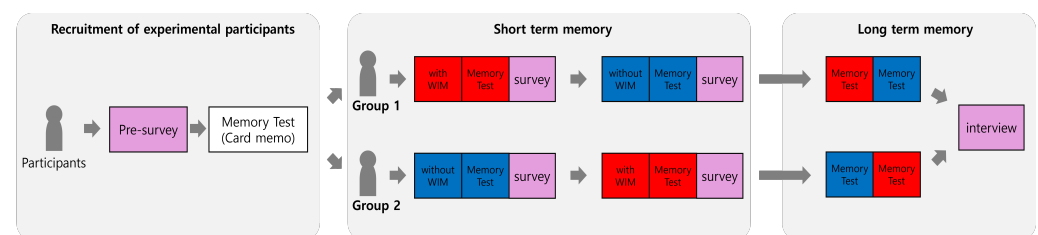


Figure 4. Experimental procedure.

Short-term memory evaluation: When participants first entered the virtual memory palace, they were required to find and memorize the images placed on the walls. These images were distributed across multiple rooms, requiring participants to search for them. The structure of the virtual world resembled a game dungeon, with multiple branching paths. As a result, if participants did not pay close attention while navigating, they may have ended up revisiting rooms they had already passed.

Participants in Group 1, assigned to first experience the virtual memory palace with the WIM, underwent the short-term memory evaluation immediately after completing the experience. In this evaluation, participants observed photos presented one by one in sequence, with a total of 60 photos to evaluate. They were asked to determine whether each photo, according to their memory, had been included in the virtual memory palace or not. If so, they selected 'True'; otherwise, they selected 'False'. After the evaluation, participants completed a survey and took a rest. Following the break, they experienced the virtual memory palace without WIM. After completing this second experience, participants underwent another short-term memory evaluation and completed a final survey. For Group 2, the sequence was reversed regarding the inclusion of WIM (Figure 5).

Long-term memory evaluation: After completing the virtual memory palace experience and short-term memory evaluations, participants returned approximately 1 to 2 weeks later for a long-term memory evaluations. During this session, participants did not re-

experience the virtual memory palace but proceeded directly to the memory evaluation. The long-term memory evaluation, like the short-term evaluation, was divided into assessments with and without WIM.

The datasets used for the long-term evaluation were the same as those used in the short-term evaluation. For instance, datasets A/C and B/D, which participants had memorized during the short-term evaluation, were reshuffled internally to test recognition. This process ensured that the memorized data from the short-term evaluation were directly reused for the long-term evaluation, maintaining consistency while testing memory retention.

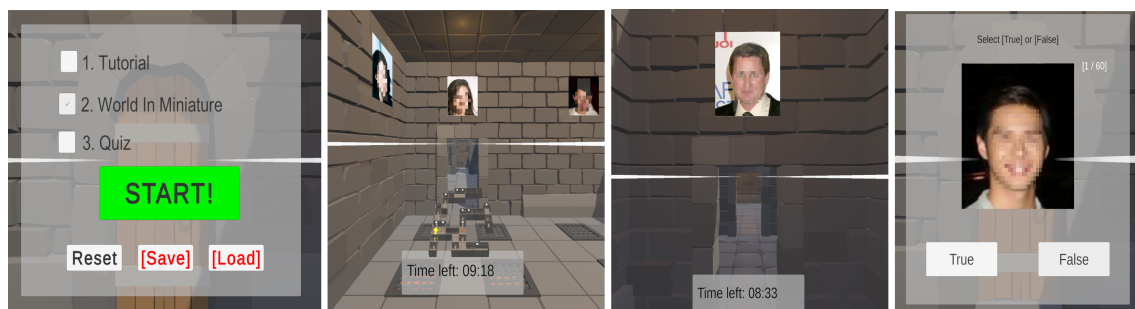


Figure 5. Procedures of our experiment in the new virtual memory palace with Worlds-in-Miniature. (The faces of the characters have been blurred to prevent infringement of portrait rights).

4.6. Measures

This section describes the types and methods of data collection used for evaluation. The data are categorized into objective data and subjective data.

Objective data include participants' memory performance evaluation data, consisting of short-term and long-term memory assessments collected after completing experiences with and without WIM. These data evaluate whether participants have correctly identified images included in the virtual memory palace, using a total of 60 images. The results are recorded as percentages, and this evaluation serves as a means to determine whether including WIM in the virtual memory palace improved memory performance or not. The data are stored after converting each participant's evaluation score into a percentile. The stored data are then analyzed using a forgetting curve model graph based on both short-term and long-term memory assessments.

A graph was created using a method similar to the forgetting curve introduced by Hermann Ebbinghaus [31]. The forgetting curve provides a model for the exponential decay of memory retention over time. This model can be expressed as follows:

$$R(t) = R_0 \cdot e^{-kt} \quad (1)$$

where $R(t)$ is the memory retention at time t , R_0 is the initial retention, and k is the forgetting rate constant. To understand the relationship between short-term and long-term memory, we define the average forgetting rate s over a specific time interval $[0, t]$ as follows:

$$s = \frac{M_{\text{short}} - M_{\text{long}}}{t} \quad (2)$$

where M_{short} and M_{long} represent the measured retention rates at short-term and long-term intervals, respectively. To connect the average forgetting rate s with the forgetting curve model, we rewrite the retention difference, as follows:

$$M_{\text{short}} - M_{\text{long}} = R_0 - R_0 \cdot e^{-kt} = R_0(1 - e^{-kt}) \quad (3)$$

Substituting this into the definition of s , we obtain the following:

$$s = \frac{M_{\text{short}} - M_{\text{long}}}{t} = \frac{R_0(1 - e^{-kt})}{t} \quad (4)$$

This equation shows that s is influenced by both the forgetting rate constant k and the time interval t . The parameter s represents the average forgetting rate over the interval $[0, t]$, while k characterizes the instantaneous forgetting rate in the Ebbinghaus forgetting curve. As $t \rightarrow 0$, s approaches k , making s a useful approximation of k over short intervals. This relationship provides a quantitative basis for comparing short-term and long-term memory retention within our experimental framework.

By calculating s in relation to time (days) with short-term and long-term memory data, the curve can be generated. The relatively gentle slope of the curve suggests that the memory evaluation method used in this experiment employed a lenient image identification approach, resulting in slower rates of forgetting.

Subjective data were collected through surveys and interviews. The surveys included the NASA-Task Load Index (NASA-TLX) 1998 (year) [32] and the Virtual Reality Sickness Questionnaire (VRSQ) [33], for experiments both with and without WIM. The Post-Study System Usability Questionnaire (PSSUQ) 3 (version) was additionally administered, but only for experiments with WIM [34].

4.6.1. NASA-TLX

The NASA-TLX allows participants to subjectively evaluate various types of load that may occur while performing tasks in a given environment. There are six evaluation items, each corresponding to the following questions. In the questions listed below, “the task given through this system” refers to exploring the given virtual space to memorize objects. The questionnaire uses a 0 (very low) to 10 (very high) scale, with lower scores indicating less effort or lower task load. Notably, for Q5 (Performance), higher scores are considered better, reflecting successful task completion, as shown in Table 1.

Table 1. NASA-TLX questionnaire table.

No.	Factor	Question
Q1	Mental	How much mental or cognitive effort is required to perform the given task using this system?
Q2	Physical	How much physical effort is required to perform the given task using this system?
Q3	Temporal	How much time pressure do you feel while performing the given task using this system?
Q4	Effort	How much overall effort do you think is needed to perform the given task using this system?
Q5	Performance	How successfully do you think you can complete the given task using this system?
Q6	Frustration	How much frustration do you feel while performing the given task using this system?

4.6.2. VRSQ

The VRSQ is a specialized assessment tool designed to measure symptoms of motion sickness or discomfort experienced by users during or after exposure to virtual reality environments. VRSQ focuses on evaluating various symptoms commonly associated with VR sickness, such as nausea, dizziness, headaches, and oculomotor strain. Unlike general-

purpose motion sickness assessments, VRSQ is tailored specifically to the unique sensory and perceptual experiences caused by VR systems. The questionnaire uses a 4-point Likert scale ranging from 0 (Not at all) to 3 (Severe), with lower scores indicating fewer symptoms of discomfort, such as reduced motion sickness, shown in Table 2.

Table 2. VRSQ questionnaire table.

Category	Factor
Oculomotor	Eye strain, blurred vision, and difficulty focusing.
Disorientation	Sensations of imbalance, spinning, or vertigo.

4.6.3. PSSUQ

The Post-Study System Usability Questionnaire (PSSUQ) is a survey designed to evaluate the usability of the proposed system after use. In this context, 'the system' refers to the virtual memory palace with WIM. Participants were asked to assess the overall usability of the system. However, some of the 16 evaluation items in the PSSUQ that were not directly relevant to our system were excluded from the survey. The evaluation used a 7-point Likert scale, ranging from 1 (very good) to 7 (very poor), to gather participants' subjective opinions. shown in Table 3.

Table 3. PSSUQ questionnaire table.

No.	Question
Q1	I am satisfied with how easy it is to use this system overall. (System: Worlds-In-Miniature, referring to the map aiding spatial understanding.)
Q2	I found this system simple to use.
Q3	I was able to complete tasks and scenarios using this system.
Q4	I felt comfortable using this system.
Q5	I found it easy to learn how to use this system.
Q6	I was able to effectively achieve the given goal (memorizing targets) using this system.
Q7	The information provided with this system (e.g., online help, screen messages, and other documentation) was clear.
Q8	The information helped me complete tasks and scenarios.
Q9	The organization of information on the system screen was clear.
Q10	I liked using this system's interface.
Q11	This system has all the functionality and capabilities I expected.
Q12	Overall, I am satisfied with this system.

4.6.4. Interview

The interviews were conducted after the long-term memory assessment. Participants were asked a simple question: "Was WIM helpful for memory recall?" Using a scale from 1 (very helpful) to 5 (not helpful), the average score was 2.1, indicating an overall positive evaluation.

The interview question encouraged participants to share their impressions of the experiment freely and without restrictions. Specifically, the responses to the prompt "Please describe your experience as a participant" were categorized and summarized according to the following two key aspects of the study: memory performance and usability.

5. Results

This section presents the analysis results of the collected data. In the objective evaluation, the short-term and long-term memory performance data collected from participants through observed responses during the evaluations were analyzed. The data were segmented into various group classifications to evaluate the significance of the experiment from multiple perspectives. In the subjective evaluation, the current usefulness and future potential of the proposed system were predicted based on the participants' responses to the questionnaires.

5.1. Objective Evaluation

In our experiment, the objective evaluation was based on the memory assessments conducted by participants after completing the experiment. Short-term memory evaluation was conducted immediately after the memorization process, while long-term memory evaluation was conducted during follow-up visits 1 to 2 weeks later.

5.1.1. Short-Term Memory

In the short-term memory assessment, participants using WIM achieved a higher identification rate of 76.1%, compared to 69.6% without WIM (Figure 6). The *t*-test (Welch's method, *t*-statistic = 1.667, *p* = 0.0998) did not show statistical significance but suggested a positive trend favoring WIM. These results align with hypothesis H1, which suggests that WIM can improve short-term memory performance by reducing cognitive load and providing immediate spatial cues.

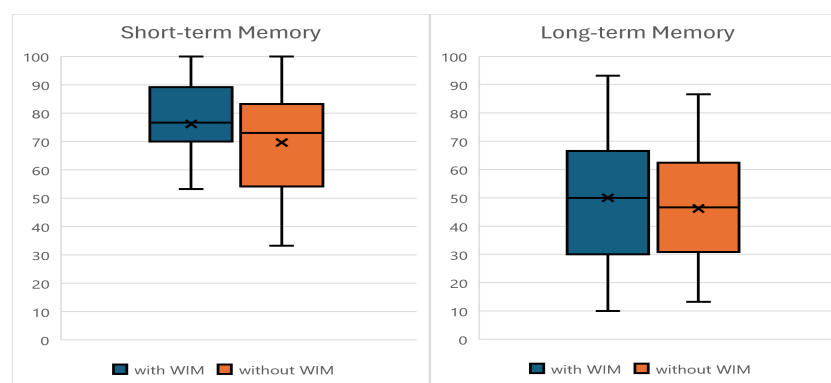


Figure 6. Short-term memory performances with and without WIM (left) and long-term memory performances with and without WIM (right).

5.1.2. Long-Term Memory

The results of the long-term memory evaluation were analyzed. First, the participants' long-term memory data, averaged without considering the time differences for re-evaluation, resulted in an average of 47.5% memorization. The experiment with WIM resulted in 50% memorization, while the experiment without WIM resulted in 46.2% memorization. Although the difference was relatively little, the experiment with WIM condition was slightly superior, as suggested by our hypothesis H1.

In the long-term memory evaluation, the effect of WIM was less pronounced. The average retention rates were 50% with WIM and 46.2% without WIM. The *t*-test results (Welch's method, *t*-statistic = 0.276, *p* = 0.783) revealed no statistically significant difference between these two conditions. This suggests that while WIM facilitates immediate spatial cognition, its benefits for long-term memory retention are relatively limited (see Figure 7). This partially supports hypothesis H2, as the marginal improvement in long-term memory suggests some benefits of WIM, though not statistically significant.

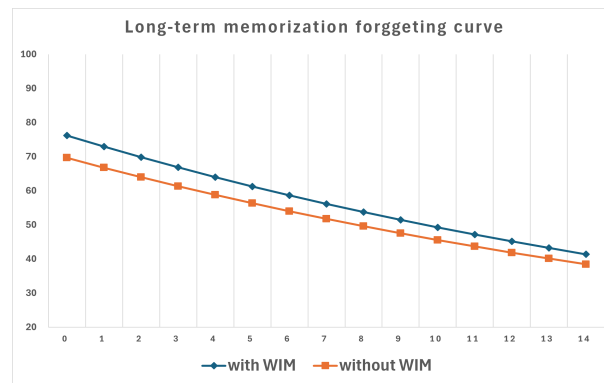


Figure 7. Estimated graphs of long-term memory decay based on the equation of the Ebbinghaus forgetting curve.

5.1.3. Superior vs. Inferior Memorization

To analyze the experimental results in greater depth, the entire participants were categorized into Above Average and Below Average groups, based on their memory retention ability. We first calculated the overall averages of memory retention scores for both the with-WIM and without-WIM conditions. If each participant's score in either the with-WIM or without-WIM condition exceeded the corresponding overall average, they were classified as part of the Above Average group. Conversely, if both scores were below the corresponding averages, they were classified as part of the Below Average group. shown in Table 4.

Table 4. *t*-test results for memory evaluations across groups and assessments.

Assessment	Group	<i>t</i> -Test Result (<i>p</i> -Value)
Short-term	Above Average	0.091723
	Below Average	0.039282
7-day assessment	Above Average	0.537406
	Below Average	0.154023
14-day assessment	Above Average	0.610577
	Below Average	0.813372

In the short-term memory evaluation, the Below Average group exhibited a statistically significant difference between the with-WIM and without-WIM conditions ($p = 0.039282$). This result highlights the effectiveness of WIM in assisting participants with lower memory retention. The observed difference suggests that WIM provides a clear advantage for this group, likely by supporting spatial understanding and aiding memory recall during the task.

In contrast, the Above Average group did not show a statistically significant difference ($p = 0.091723$), although there was a slight trend favoring the WIM condition. This indicates that participants with higher memory retention may rely less on external aids like WIM to perform well in short-term memory tasks.

These results highlight that WIM can significantly benefit people with relatively weak memory skills. Specifically, WIM simplifies the task of navigation by offering a miniature overview of the environment, which minimizes the mental effort required to process and recall spatial information. This reduces cognitive load, allowing participants to focus more on encoding task-relevant images into memory. Additionally, WIM's consistent and visual spatial representation mitigates the risk of disorientation, preventing unnecessary backtracking or re-navigation, which often disrupt memory encoding processes.

The implications of these findings are significant for individuals with lower memory retention. WIM demonstrates its potential as an assistive tool, particularly in contexts where spatial navigation and memory recall are challenging. For educational or training environments, WIM could be integrated into virtual systems to enhance the learning experience for participants who face cognitive challenges. Its ability to reduce mental effort and to enhance immediate task performance underscores its value as a supportive technology for real-time tasks. However, for long-term memory, particularly in the Below Average group, the without-WIM condition showed better retention, underscoring WIM's limitations for extended memory retention and suggesting that complementary strategies may be necessary to address this gap. These findings strongly support hypothesis H3, demonstrating that WIM is particularly beneficial for participants with lower baseline memory performance by reducing cognitive load and enhancing spatial understanding. (see Figure 8).

5.1.4. Gaming and VR Experience Analysis

To assess whether prior experience with 3D games and VR influenced participants' memory performance, an additional analysis was conducted. Participants' self-reported gaming and VR experience scores (collected via pre-study surveys) were used as variables. The correlation between these variables and memory performance was analyzed using statistical tests (Pearson correlation and *t*-tests).

The results showed no statistically significant relationship between gaming/VR experience and memory retention in either short-term or long-term evaluations ($p > 0.1$). This suggests that prior experience with digital or virtual environments did not strongly affect participants' ability to utilize the virtual memory palace. However, future research could explore whether larger sample sizes or more specific categories of experience reveal different outcomes.

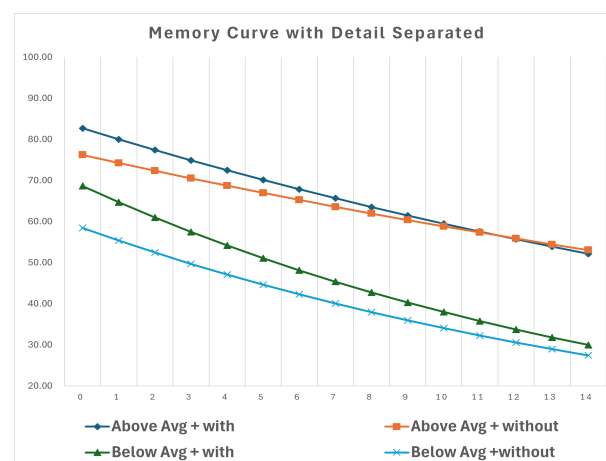


Figure 8. Long-term memory forgetting curves for 4 different conditions, created by combining 2 different properties (namely, above/below average memory ability, with/without WIM).

5.2. Subjective Evaluation

This section presents the analysis results of the collected subjective data. In the subjective evaluation, the usefulness of the proposed system was reviewed based on participants' survey responses, including responses to the NASA-TLX, VRSQ, PSSUQ, and the interview.

5.2.1. NASA-TLX

The results of the NASA-TLX survey can be seen in Figure 9. The left side represents the data evaluated after the experiment with WIM, while the right side shows the data

evaluated after the experiment without WIM. In this evaluation, lower scores indicate less perceived burden for participants, except for Q5 (Performance). Comparing the two graphs, the WIM condition shows lower average scores in Q1 (mental demand), Q2 (physical demand), Q3 (temporal demand), Q4 (effort), and Q6 (frustration), indicating that WIM effectively reduced the overall burden experienced by participants.

To verify statistical significance, a *t*-test was conducted for each item. Statistically significant differences were observed in mental effort (Q1), overall effort (Q4), and performance satisfaction (Q5), while no significant results were found for physical effort (Q2), time pressure (Q3), or frustration (Q6). These results suggest that, despite similar levels of physical load, time burden, and frustration, participants experienced reduced mental effort and greater performance satisfaction. These findings further support hypothesis H1, as reduced cognitive load and improved task performance align with the expectation that WIM facilitates memory tasks by simplifying spatial navigation.

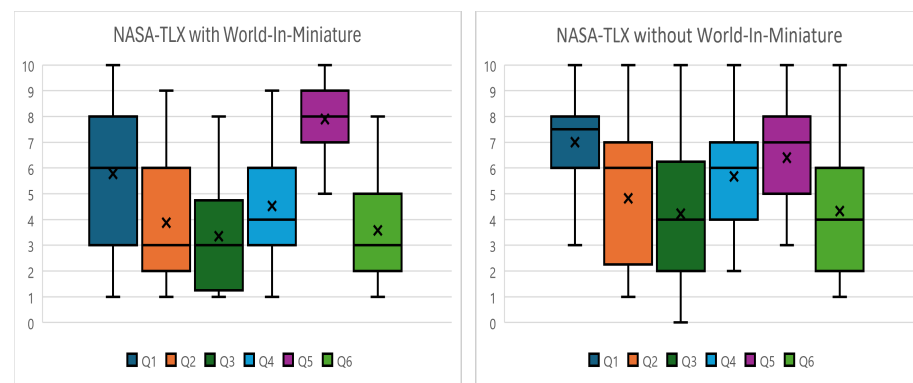


Figure 9. Graphs summarizing the responses to NASA-TLX questionnaire for with-WIM (left) and without-WIM (right) conditions.

5.2.2. VRSQ

The VRSQ is a questionnaire designed to allow participants to subjectively evaluate the motion sickness induction index in VR HMD content. It evaluates two major aspects, one concerning eye movement and the other concerning disorientation. As shown in Table 5, the motion sickness induction index showed significant differences of 9.05.

A *t*-test was conducted on the classification of eye movement and disorientation for the two groups with and without WIM. The results showed that eye movement was not statistically significant at $p = 0.1116$. However, disorientation was statistically significant at $p = 0.0489$, indicating that the spatial cognitive information provided by WIM was effective in reducing disorientation.

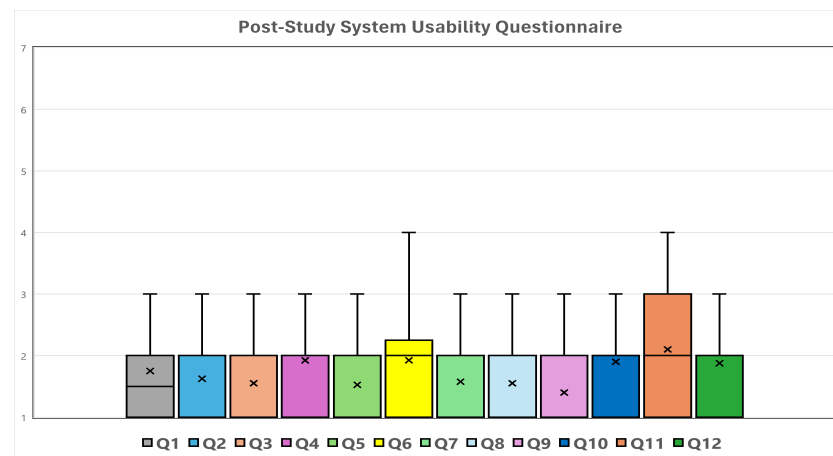
For the total sickness score, a *t*-test was directly conducted to compare the two groups with and without WIM. The results showed that $p = 0.0544$. While this value is slightly above the conventional threshold for statistical significance ($p < 0.05$), it suggests a trend toward significance. This means that WIM may have had a modest positive effect on reducing motion sickness overall, though the evidence is not strong enough to conclusively reject the null hypothesis. Therefore, while not strongly significant, the results provide preliminary support for the idea that WIM can improve user comfort in VR environments by mitigating motion sickness. These results support hypothesis H1, as improved spatial orientation through WIM may contribute to reduced cognitive and physical discomfort, enhancing short-term task performance.

Table 5. VRSQ evaluation results according to the experiments with and without WIM and their differences.

State	Oculomotor	Disorientation	Sickness
With WIM	31.9	25.7	28.8
Without WIM	39.2	36.5	37.8
Difference	7.23	10.83	9.06

5.2.3. PSSUQ

The collected evaluation results can be seen in Figure 10. Overall, the low scores indicate that WIM included in the virtual memory palace was helpful to users during the memorization tasks. However, Q6 and Q11 showed noticeably higher averages and standard deviations compared to other items, indicating greater variability in user responses. This discrepancy can likely be attributed to challenges such as the limited intuitiveness of the WIM interface for achieving specific tasks (e.g., identifying memorization targets quickly) and inadequate guidance provided by the system on its functionality and usage scenarios. Further insights into these issues, including user feedback on Q6 and Q11, are discussed in the following Section 5.2.4. The usability benefits align with the hypothesis H3 by demonstrating that participants with lower baseline performance found WIM beneficial for reducing task complexity and improving memory recall.

**Figure 10.** Evaluation results summarizing the responses to PSSUQ.

5.2.4. Interview

The average score from the interview responses was 2.1, suggesting that WIM was generally helpful. The final open-ended question allowed participants to freely share their thoughts, which were categorized into the following two main aspects: memory performance and usability.

Memory performance. Participants provided overall positive feedback regarding memory performance. Some mentioned that “With WIM, it was easy to navigate the world and optimize the process of memorizing paths, but applying WIM to more complex processes in short-term and long-term memory might have shown clearer differences”. Another said, “The memory palace was effective for memorization, and the inclusion of a miniature helped significantly. Usually, I am not good at remembering faces, but the miniature memory palace allowed me to remember much better”. Others noted, “I was able to remember for a long time with WIM”, and added, “It was fascinating that I could recall so much even after a week”. One participant remarked, “The world-in-miniature was more suitable not only for memory performance but also for gaming and utilization compared to

the typical 2D format". Some participants observed, "Without WIM, I had to concentrate more on understanding the map, which helped me retain information better". Another key observation was, "Although using WIM showed lower immediate performance, it enhanced retention in the long term". Particularly, one participant highlighted, "In the second experiment conducted a week later, I was able to clearly remember who was in the room with WIM, whereas without WIM, I got confused between participants from experiments A and B". Overall, the participants concluded, "WIM enhanced spatial memory and left a more detailed image in my mind".

However, some participants provided negative feedback regarding memory performance. One participant commented, "I am not sure if WIM was really helpful for memory recall". Another said, "Although I used the memory palace, I am uncertain if it was effective for memorization". Others noted, "Without using WIM, I was able to recheck places or avoid going to the ends of the map, which made my memory more efficient". These insights suggest that the efficacy of WIM needs further verification.

Usability: There was generally positive feedback given regarding usability. Participants remarked, "The experiment using WIM was very interesting, as it resembled an image-scanning method for studying". Another said, "Although there was VR motion sickness, it was fun to experience the differences while matching pictures". One participant noted, "Having a miniature significantly improved the efficiency of navigation and spatial memory, which made the user experience very convenient". Others observed, "I was able to directly experience how spatial cognition affects memory". Some participants suggested, "Having a large object in the room might have further aided memory recall". Another remarked, "Using VR for memorization was refreshing and more effective than traditional methods".

Despite the generally positive feedback, some usability issues were identified. A few participants noted, "WIM was helpful for locating the current position, but I am not sure if it actually improved memory recall". Another said, "Without WIM, it was much more comfortable". These responses highlight the need for further refinement of WIM's interface to better meet user expectations.

The usability-related feedback partially aligns with the PSSUQ results discussed in Section 5.2.3. Questions about the system's integration of functions and the effort required to use the system showed higher variability in participant responses. These findings suggest that challenges such as ambiguous room distinctions and the fixed position of WIM affected usability. Addressing these issues, such as by improving room clarity and providing adjustable features for WIM, could significantly enhance its usability and effectiveness in memory-related tasks.

Participants generally reported that WIM was helpful for spatial navigation and memory recall, though some participants expressed doubts about its long-term effectiveness. The feedback revealed that WIM improved immediate spatial cognition and task performance, aligning with hypothesis H1. However, the mixed feedback on long-term memory aligns with the partial support for hypothesis H2, suggesting that additional strategies may be necessary to achieve significant long-term retention improvements.

6. Discussion

This section interprets the results in relation to the proposed hypotheses and discusses their implications for memory performance and spatial navigation.

6.1. Hypothesis 1: WIM Enhances Short-Term Memory

The positive trend in short-term memory results (76.1% with WIM vs. 69.6% without WIM) and the reduced cognitive load observed in the NASA-TLX evaluation sup-

port hypothesis H1. Although the statistical significance threshold was not reached ($p = 0.0998$), the proximity to this threshold suggests that WIM effectively aids memory recall by providing immediate spatial context. The reduced mental demand and effort reported in the subjective evaluations reinforce this finding. Future studies with larger sample sizes could help validate and strengthen this evidence.

6.2. Hypothesis 2: WIM Enhances Long-Term Memory Retention

Although the long-term memory evaluation showed slightly higher retention rates with WIM (50%) compared to without WIM (46.2%), the difference was not statistically significant ($p = 0.783$). This partial support for hypothesis H2 indicates that WIM's benefits for long-term retention may be limited. Interviews and subjective feedback further suggested that while WIM enhances immediate recall, its effect on long-term retention is less pronounced. Previous studies have shown that spatial scaffolds, such as distinctive spatial contexts, can enhance memory retention by providing robust cues for recall [20]. However, our findings suggest that, while WIM provided immediate spatial context, this effect diminished over time. This aligns with previous research indicating that for long-term memory improvements, spatial scaffolds alone may be insufficient unless supplemented by repetitive learning or interactive review mechanisms [7,20]. Incorporating complementary strategies, such as spaced repetition or interactive review sessions, could enhance long-term memory retention in future implementations.

6.3. Hypothesis 3: WIM Is More Beneficial for Participants with Lower Memory Performance

The significant improvement in short-term memory retention among the Below Average group ($p = 0.039$) strongly supports hypothesis H3. Participants with lower baseline memory retention benefited from WIM's ability to reduce cognitive load and improve spatial orientation. The subjective evaluations, particularly the PSSUQ and interview responses, further confirmed that WIM simplified navigation and enhanced task performance for this group. In contrast, participants in the Above Average group did not show significant improvement, indicating that they may rely more on internal memory strategies and less on external aids like WIM.

6.4. Limitations and Future Work

While the results provide valuable insights, several limitations should be addressed. The small sample size may have limited the ability to detect statistically significant effects, particularly in the long-term memory evaluations. Additionally, the study focused on a specific virtual environment and task design, which may limit its generalizability. Future research could explore diverse environments, larger participant groups, and additional complementary memory techniques to enhance long-term retention. Given the effectiveness of distinctive virtual environments in improving memory performance [6,21], future work could focus on incorporating more complex and realistic spatial contexts to enhance long-term retention. Distinctive landmarks, dynamic environmental interactions, or multi-sensory cues could help reinforce memory traces and support robust retrieval across extended periods. Although participants had access to the teleportation function during the experiment, quantitative data on its usage frequency was not collected. Since teleportation can significantly influence the path traversal and spatial memory encoding process in the method of loci, future studies should investigate its impact more systematically by tracking and analyzing its usage.

7. Conclusions

This study investigated the integration of a virtual memory palace with Worlds-in-Miniature (WIM) to enhance spatial cognition and memory performance in virtual reality

environments. The motivation stemmed from the limitations of traditional memory palaces, which can be inaccessible to untrained users and challenging to implement in virtual spaces. By introducing procedural generation methods and an intuitive spatial interface, we aimed to address these limitations.

Our findings demonstrate that WIM significantly improves short-term memory performance, particularly for participants with lower baseline memory abilities. The global spatial representation provided by WIM allowed these participants to better structure and recall information, as confirmed by both objective and subjective evaluations. However, WIM's benefits were less pronounced for long-term memory, as retention rates between conditions converged over time. This suggests that while WIM facilitates immediate task performance, additional mechanisms, such as repeated learning or reinforcement strategies, may be necessary to support long-term retention.

Subjective evaluations further supported these findings, with participants reporting reduced mental effort and disorientation when using WIM, which translated into improved task efficiency. However, some participants highlighted limitations regarding WIM's fixed size and placement, indicating the need for customizable interface options to better suit individual preferences and task requirements.

Despite its promising results, this study has certain limitations. The procedurally generated virtual environments lacked the realism and semantic cues often present in traditional memory palaces, which may have influenced memory performance. Additionally, the relatively short evaluation periods and limited sample size constrain the generalizability of the results. Future research should address these limitations by incorporating more complex and realistic spatial layouts, longer observation periods, and larger, more diverse participant groups.

In conclusion, the integration of WIM into virtual memory palaces effectively enhances short-term memory performance and provides substantial support for individuals with weaker memory retention. Moving forward, improvements such as customizable interfaces, advanced spatial cues, and repeated exposure mechanisms could further optimize the system for use in educational and industrial applications requiring efficient memorization and navigation.

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Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of Kwangwoon University (7001546-202400830-HR(SB)-008-02).

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The data are available upon request due to privacy or other restrictions.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. How to Implement a Virtual Memory Palace

This section explains our new approach for generating a virtual memory palace that is equipped with WIM, which is used as a tool to enhance both spatial perception and navigation within the virtual memory palace.

Appendix A.1. System Overview

One characteristic that differentiates our proposed virtual memory palace from traditional virtual palaces is that it features a dynamically generated world created instantaneously. Additionally, our memory palace provides users with WIM, allowing them to observe the entire world while gaining objective information about their position and orientation. The creation of the world leverages procedural generation techniques to provide the virtual memory palace with diverse structures a predefined set of parameters. The enhancement of visual information through WIM aims to enable immediate usability of the virtual memory palace without requiring prior construction of the world or exploratory activities to build familiarity, unlike traditional memory palaces.

Our system was developed on a Windows 11 environment with an i7-10700K processor, RTX-2080s GPU, and 32 GB of memory. The Unity engine, version '2021.3.16f', was used to develop the virtual reality environment. The connection between the engine and VR was established using the 'XR Interaction Toolkit 2.2.0'. The VR HMD hardware utilized was the 'Meta Quest 3'.

Appendix A.2. Procedural Generation of Memory Palaces

Our virtual world was based on a flat plane, capable of accommodating rooms of various sizes in a grid. Larger rooms were placed first and smaller rooms were placed later in empty spaces to prevent overcrowding and maintain a minimum gap. Each room was connected to its adjacent room via a path. Regarding the rooms and paths as the nodes and edges, the entire world could be effectively encoded as a graph structure. Applying the Prim's Minimum Spanning Tree algorithm to this graph ensured that all rooms were linked to form a connected component. Finally, each cell in the grid was represented in 3D, with walls and open passages constructed accordingly. This process is explained in detail below (see Figure A1).

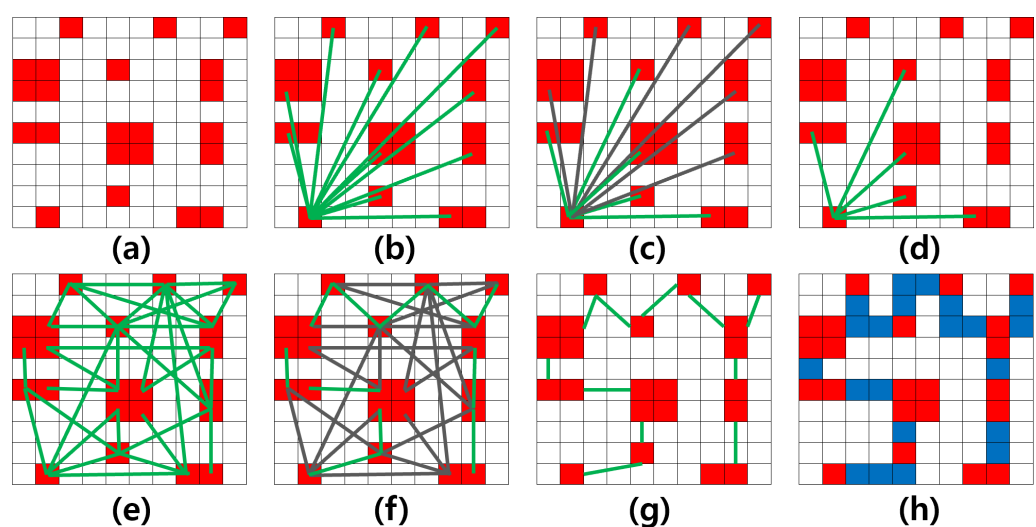


Figure A1. Procedural generation process of a virtual memory palace. Tiles colored red and blue are rooms and paths, respectively. Line segments colored light green and dark green are all possible rays to be added to a graph and some intersected rays among those to be removed, respectively, in the search process.

First, to simplify and expedite the creation process of the world, we limited it to a single-level, flat plane without elevation. The world was represented as a two-dimensional $N \times N$ grid, which we referred to as the *tile map*. Each tile in the tile map was assigned one of three attributes: *Room* (representing a room), *Path* (representing a path), or *None* (representing an empty space). The minimum size of each tile was 1×1 , corresponding to approximately 4 m per side in real-world dimensions. This size was chosen to ensure collision safety and allow free movement within the physical space used for the experiment. Depending on the room's area and the placement of objects, the size could be adjusted flexibly. Room sizes ranged from 1×1 to $M \times M$, where M must be significantly smaller than N to ensure sufficient space for additional rooms and paths. In our experiment, M was limited to 20% of N , which was determined in order to provide a balance between spatial diversity and efficient use of the available tile map, based on several trial simulations conducted during the experiment design process.

Second, the list of rooms was sorted in descending order of size, determining which room was created first. The position of each room was assigned a random index between 0 and $N - 1$. The first room could be placed anywhere on the tile map. However, for subsequent rooms, the placement had to consider the existing rooms. If a collision occurred due to the position or size of a newly placed room, the placement process was repeated until no collisions occur. Additionally, to prevent rooms from directly touching each other, a minimum gap around each tile was maintained on the tile map (see Figure A1a).

The third step involved generating paths to connect the rooms. Rays were cast from the center of each room to all other rooms (Figure A1b). Each ray was associated with its starting and ending room. Among these rays, any ray that intersected a room other than its starting or ending room was removed (Figure A1c,d). Once all rooms and rays were connected, the result formed a graph structure (Figure A1e). In this graph, each room acted as a node, and each ray served as an edge. To ensure all rooms were connected, the Prim's Minimum Spanning Tree algorithm was applied to the graph (Figure A1f). After connecting all rooms, the rays were rasterized onto the tile map (Figure A1g). This process involved iterating through all tiles and converting tiles intersected by rays into paths (Figure A1h).

In our virtual memory palace, images for memorization could be placed and arranged according to the experimental requirements. A single-unit room was able to display a maximum of four images, determined by the number of walls that allowed for image placement. However, to reduce visual density, we limited the number of images to half the available walls. Rooms of the smallest size (1×1) had four walls, but only two were used for displaying images. Rooms with two units had six walls, allowing for three images, while rooms with three units had eight walls, allowing for four images. Images were randomly assigned and placed based on the number allocated to each room (see Figure A2). This system could accommodate an increased number of images depending on the number of rooms or memorization targets. This means that the system is not limited to the current experimental conditions and can be scaled up to larger environments or more complex structures as needed. For instance, more rooms and images can be added, or a more intricate spatial layout can be designed to adapt the system to various experimental purposes. However, for the experiment, we consistently fixed the number of allocated images to 30 in order to reduce the cognitive load on participants and keep the overall task manageable.

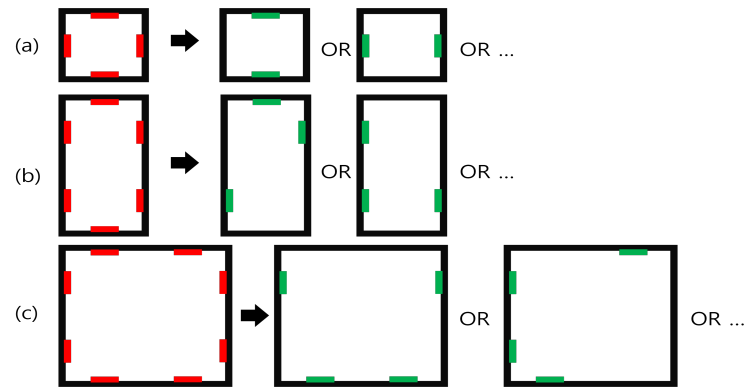


Figure A2. Arrangement of memorization target images in the room. Red segments indicate all possible positions where images can be placed, while green ones highlight half of those positions that are randomly selected. (a) The smallest size room (one unit) has four walls, with two walls randomly selected for image placement. (b) A two-unit room has six walls, with three walls used for image placement. (c) A four-unit room has eight walls, with four walls used for image placement.

Algorithm A1 Procedural World Building

Require: H, W ▷ Height, Width of the map
Require: *RoomPlacingDataList* ▷ List of (u, v) info for each room

```

1: TileMap ← CREATETILES( $H, W$ )
2: INITIALIZE(TileMap, "None")
3: RoomList ← ∅

4: for  $i$  ← 0 to COUNT(RoomPlacingDataList) − 1 do
5:   ( $u, v$ ) ← RoomPlacingDataList[ $i$ ]
6:   ( $x, y$ ) ← GETRANDOMPOSITION( $H, W$ )
7:   room ← CREATEROOM( $x, y, u, v$ )
8:   isIntersect ← False
9:   for all  $r$  in RoomList do
10:    if INTERSECT(room,  $r$ ) then
11:      isIntersect ← True
12:      break
13:   if not isIntersect then
14:     ADDROOM(RoomList, room)
15:     for  $r$  ← room.y to room.y + room.v − 1 do
16:       for  $c$  ← room.x to room.x + room.u − 1 do
17:         TileMap[ $r$ ][ $c$ ] ← "Room"

18: RayList ← ∅
19: for all  $room_1$  in RoomList do
20:   for all  $room_2$  in RoomList do
21:     if  $room_1 \neq room_2$  then
22:       ray ← RAYCAST( $room_1, room_2$ )
23:       ADDRAY(RayList, ray)
24: filteredRays ← FILTERWITHOUTROOMCOLLISION(RayList)
25: MST ← GENERATEMINIMUMSPANNINGTREE(filteredRays)
26: for all connection in MST do
27:   ray ← connection.ray
28:   for all tile in GETTILESALONGRAY(TileMap, ray) do
29:     if TileMap[tile.x][tile.y] = "None" then
30:       TileMap[tile.x][tile.y] ← "Path"

```

▷ — Build Rooms —

▷ — Build Paths —

Algorithm A2 Generate Minimum Spanning Tree (Prim's Algorithm)**Require:** *edges*: List of edges (*Room*₁, *Room*₂, *Ray*_{start}, *Ray*_{end})**Ensure:** *mstEdges*: List of edges forming the MST

```

1: mstSet  $\leftarrow \emptyset$  ▷ Set of rooms in the MST
2: mstEdges  $\leftarrow \emptyset$  ▷ List of edges in the MST
3: if COUNT(edges) = 0 then return mstEdges
4: ADD(mstSet, edges[0].Item1) ▷ Add the first room to the MST
5: while COUNT(mstSet) < COUNT(rooms) do
6:   minWeight  $\leftarrow \infty$ 
7:   minEdge  $\leftarrow$  null
8:   for all edge  $\in$  edges do
9:     if edge.Item1  $\in$  mstSet and edge.Item2  $\notin$  mstSet then
10:      weight  $\leftarrow$  DISTANCE(edge.Item3, edge.Item4)
11:      if weight < minWeight then
12:        minWeight  $\leftarrow$  weight
13:        minEdge  $\leftarrow$  edge
14:      else if edge.Item2  $\in$  mstSet and edge.Item1  $\notin$  mstSet then
15:        weight  $\leftarrow$  DISTANCE(edge.Item4, edge.Item3)
16:        if weight < minWeight then
17:          minWeight  $\leftarrow$  weight
18:          minEdge  $\leftarrow$  edge
19:   if minEdge  $\neq$  null then
20:     ADD(mstEdges, minEdge)
21:     ADD(mstSet, minEdge.Item1)
22:     ADD(mstSet, minEdge.Item2)
return mstEdges

```

Appendix A.3. Interactive Manipulation of Worlds-in-Miniature

To enhance the effectiveness of the virtual memory palace, we introduced the WIM technique. WIM serves as a powerful tool that enables users to instantly grasp the spatial structure and effortlessly recognize their position and orientation, even in unfamiliar environments. This technique is particularly valuable in complex 3D virtual environments, simplifying spatial navigation and improving memorization efficiency.

We scaled down and replicated the world generated by a tile map to create a miniature environment, keeping it synchronized with the original virtual world. This miniature remains visible within the user's field of view at all times. The user's position and orientation are depicted by an avatar, which serves as a simplified representation of the user's real-world position and orientation.

In the WIM we implemented, users can choose one of two modes. In the first mode, the orientation of the WIM remains fixed in a designated position on the screen (e.g., the bottom-right corner). When the user moves or rotates, the WIM itself does not move; instead, the avatar inside the WIM reflects the user's movement and directional changes by rotating or shifting position. For example, if the user physically turns 90 degrees, the avatar inside the WIM appears to rotate 90 degrees, while the WIM remains stationary. In contrast, in the second mode, the WIM stays aligned with the user's viewpoint. As the user rotates, the WIM itself rotates accordingly, maintaining alignment with the user's perspective, while the avatar inside the WIM always stays aligned with the user's viewpoint.

These two modes were designed to accommodate various user needs and task scenarios. The first mode provides a fixed overview of the environment, which is advantageous for understanding the structure of the space and planning exploration strategies. On the other hand, the second mode allows the WIM to stay synchronized with the user's viewpoint, making it easier for users to intuitively understand their current position and

orientation. Each mode offers unique advantages that enhance the user experience in different ways.

The first mode allows users to maintain a fixed view of the overall environment, making it easier to grasp the spatial structure as a whole. Additionally, since the orientation of the WIM does not change with the user's movements, this mode minimizes potential confusion. However, because the WIM is not aligned with the user's viewpoint, it may be less intuitive for users to perceive their direction and position. Conversely, the second mode helps users intuitively understand their current direction and position by keeping the WIM synchronized with their viewpoint. This alignment provides more immersive feedback, making it easier for users to connect their physical movements with the virtual space. However, as the WIM rotates with the user, the overall spatial structure may become harder to interpret and remember.

Additionally, users can directly interact with the WIM. An avatar, representing the user's position and orientation, is placed within the miniature, allowing users to drag and drop it to move to a specific location in the world. As soon as the drag-and-drop action occurs, the user is transported to the selected location. This method allows for relatively precise long-distance movement with less effort than conventional navigation methods.

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